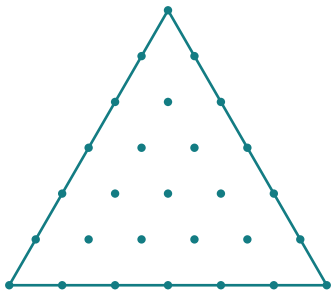
PROBABILITY AND MATHEMATICAL MODELLING

Discrete Information Geometry

Finite Differences, Probability, and Information Flows

Chapter 7



A course in ten chapters

https://bobovski66.github.io/IntroDiscreteInfoGeom/

CHAPTER 7

Discrete Fisher geometry and statistical distance

7.1 Two notions of length for a finite step

Let p and q be probability vectors. A simple coordinate distance is the Euclidean norm $\|q - p\|$. But equal coordinate changes can be more significant when they affect rare outcomes. Information geometry uses another notion of length to describe this statistical effect.

For positive p_i and a small step $\delta = q - p$ with $\sum_i \delta_i = 0$, the Fisher quadratic form is

$$\boxed{g_p(\delta, \delta) = \sum_i \frac{\delta_i^2}{p_i}.}$$

Its square root is a local approximation to the statistical length of a small move. Notice that the denominators depend on the initial state. A change to a rare outcome counts more.

The relationship with relative entropy is

$$2D(p + \delta \| p) \approx g_p(\delta, \delta)$$

for small displacements in the interior. Thus the local length is $\sqrt{g_p(\delta, \delta)}$, with no additional factor of $\sqrt{2}$. We can compare the two expressions numerically before learning the continuous Taylor expansion that explains the approximation.

7.2 An exact distance without derivatives

Square-root coordinates give every probability vector a point on the radius-two sphere:

$$s(p) = (2\sqrt{p_1}, \ldots, 2\sqrt{p_k}), \qquad \|s(p)\| = 2.$$

The exact Fisher–Rao spherical distance between categorical distributions is

$$\boxed{d_{\mathrm{FR}}(p, q) = 2 \arccos\left(\sum_i \sqrt{p_i q_i}\right).}$$

This formula makes sense even when some probabilities are zero. It is an exact finite distance, rather than the local approximation defined above.

For Bernoulli distributions $(x, 1-x)$ and $(y, 1-y)$, the formula becomes

$$d_{\mathrm{FR}}(x, y) = 2\left|\arcsin\sqrt{x} - \arcsin\sqrt{y}\right|.$$

The moves $0 \to 1/4$ and $1/4 \to 1/2$ have equal coordinate length $1/4$, but their Fisher–Rao lengths are respectively $\pi/3$ and $\pi/6$.

7.3 Give the probability lattice edge lengths

Return to the token lattice $\mathcal{P}_N$ from Week 3. Neighboring states are joined by an edge representing one token transfer. Assign that edge the exact weight

$$\ell(p, q) = d_{\mathrm{FR}}(p, q).$$

Now the lattice becomes a **weighted graph**. A path consists of finitely many legal token transfers, and its length is the sum of its edge weights.

For a short interior edge, we can also use $\ell(p, q) \approx \sqrt{\sum_i (q_i - p_i)^2 / p_i}$. Near a zero coordinate, do not use this approximation: it divides by zero, while the exact spherical distance remains finite.

7.4 A refinement experiment

Compute the exact length of one binary move from $x = 1/4$ to $x = 1/2$. Next split the interval into two moves through $x = 3/8$ and add their exact distances. Because the binary states lie along one spherical arc in order, the sum agrees exactly. Try local quadratic edge approximations instead: their total changes under refinement and gets closer to the exact arc distance as the steps become small.

This is a direct route from discrete probability geometry toward the continuous Fisher metric developed in our companion course.

7.5 Further investigations: Discrete Fisher geometry and statistical distance

The probability triangle is an excellent map of compositions, but its ordinary Euclidean distances treat all coordinate changes in the same way. Statistical comparison also depends on where the probability mass sits. Taking square roots reveals another geometry, one in which a probability table becomes a point on a sphere.

This chapter uses lengths, dot products, and angles. Angles will be measured in radians: a full turn is 2π, a right angle is $\pi/2$, and an arc of angle α on a circle of radius R has length $R\alpha$.

7.6 Square roots place probabilities on a sphere

Map a probability vector p to the vector

$$s(p) = (2\sqrt{p_1}, \ldots, 2\sqrt{p_k}).$$

Its squared length is $\sum_i (2\sqrt{p_i})^2 = 4\sum_i p_i = 4$. Thus $s(p)$ lies on a sphere of radius 2. Every coordinate is nonnegative, so only the nonnegative portion of the sphere is used. Conversely, a nonnegative vector s of length 2 gives probabilities $p_i = s_i^2/4$.

For two categories, the image is a quarter circle. For three, it is an octant of an ordinary sphere. The same formula works for more categories, although a direct spatial picture is then unavailable. Dot products and lengths still make sense as finite sums.

Example 7.1 (Pure and uniform distributions)**.** The pure three-category state $(1,0,0)$ maps to $(2,0,0)$. The uniform state maps to $(2/\sqrt{3}, 2/\sqrt{3}, 2/\sqrt{3})$. Two different pure states lie at right angles as viewed from the center of the sphere. They are maximally separated within its nonnegative portion.

7.7 Overlap determines the angle

The **Bhattacharyya coefficient**, or square-root overlap, is

$$B(p,r) = \sum_i \sqrt{p_i r_i}.$$

The dot product of the two sphere vectors is $s(p)\cdot s(r) = 4B(p,r)$. Since both lengths are 2, the cosine of their angle α is $B(p,r)$. The Cauchy–Schwarz inequality ensures $B \le 1$:

$$\left(\sum_i \sqrt{p_i r_i}\right)^2 \le \left(\sum_i p_i\right)\left(\sum_i r_i\right) = 1.$$

Also $B \ge 0$. The angle therefore lies in $[0, \pi/2]$.

We define the **categorical Fisher–Rao distance** by the length of the shorter spherical arc:

$$d(p,r) = 2\arccos B(p,r).$$

This normalization uses the sphere of radius 2. Equality of the tables gives distance zero; disjoint supports give distance π. The construction is a standard finite probability geometry. Its arc formula can be used without developing differential geometry.

Example 7.2 (A distance calculated exactly)**.** Let $p = (1,0)$ and $r = (1/4, 3/4)$. Their overlap is $1/2$, so their angle is $\pi/3$ and their distance is $2\pi/3$. The coordinate difference of the first probabilities is $3/4$; it measures a different feature from the spherical distance.

7.8 Chords, arcs, and Hellinger distance

The straight chord joining the sphere vectors has squared length

$$\|s(p) - s(r)\|^2 = 8\big(1 - B(p,r)\big).$$

To verify this, expand the square and use the squared lengths 4 and the dot product $4B$. The **Hellinger distance** with the convention used here is

$$h(p,r) = \sqrt{1 - B(p,r)} = \frac{1}{\sqrt{2}} \left(\sum_i (\sqrt{p_i} - \sqrt{r_i})^2 \right)^{1/2}.$$

Thus the chord length is $2\sqrt{2}\,h$. The arc and chord are different, but both increase as overlap decreases. Hellinger distance, spherical distance, and relative entropy should always be identified by their formulas, since their numerical scales and boundary behavior differ.

Example 7.3 (Behavior at zero probabilities)**.** For $p = (1,0)$ and $r = (0,1)$, the overlap is zero, the Hellinger distance is one, and the spherical distance is π. Both directed relative entropies are infinite. The square-root construction includes the boundary of the probability simplex at finite distance.

7.9 A complete coordinate for binary probabilities

For a binary distribution $(x, 1-x)$, put

$$\phi = \arcsin\sqrt{x}, \qquad 0 \le \phi \le \pi/2.$$

Then $x = \sin^2\phi$ and $s = (2\sin\phi, 2\cos\phi)$. The angle ϕ specifies a point on the quarter circle. Since the radius is 2, the arc coordinate measured from $x = 0$ is 2ϕ. Two binary distributions therefore have distance

$$d\big((x,1-x),(y,1-y)\big) = 2\left|\arcsin\sqrt{x} - \arcsin\sqrt{y}\right|.$$

This gives a useful distinction between equal changes in probability and equal changes in statistical distance.

Example 7.4 (Equal probability increments at different locations)**.** The distance from $x = 0$ to $x = 1/4$ is $\pi/3$. The distance from $x = 1/4$ to $x = 1/2$ is $\pi/6$. Both changes add $1/4$ to the probability, yet the first covers twice the spherical arc. Near a boundary, introducing probability into a previously absent category can be a substantial geometric change.

7.10 Why the arc formula is a distance

The shorter great-circle arc joins any two points on a sphere unless they are antipodal. Our nonnegative points are never antipodal. Elementary spherical geometry identifies this arc as a shortest spherical route. Concatenating a route from p to q with one from q to r gives a route from p to r, whose length cannot be smaller than the shortest route. Hence

$$d(p,r) \le d(p,q) + d(q,r).$$

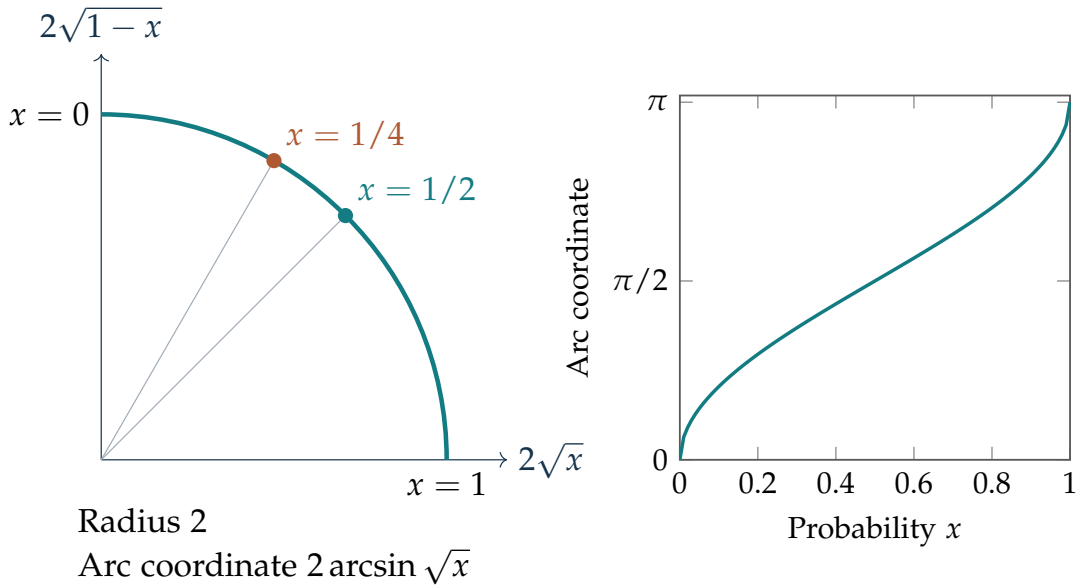


Figure 7.1. Binary distributions form a quarter circle on the radius-two sphere. Equal probability increments generally cover unequal arc lengths.

This is the triangle inequality. Symmetry follows from symmetry of overlap, and zero distance forces equal square-root coordinates, hence equal probabilities. These properties make d a distance. In Chapter 8 we will construct intermediate points on the shortest arc using only trigonometry.

When calculations are performed on a computer, tiny rounding errors can push an overlap slightly above one. Before applying inverse cosine, clamp the computed overlap to $[0, 1]$. Exact mathematics already guarantees this interval; the adjustment handles floating-point arithmetic rather than changing the definition.

7.11 A sum-of-squares proof of the overlap bound

The inequality used to bound overlap can be established directly. For any two lists a_i, b_i, expansion gives the identity

$$\left(\sum_i a_i^2\right)\left(\sum_i b_i^2\right) - \left(\sum_i a_i b_i\right)^2 = \sum_{i<j} (a_i b_j - a_j b_i)^2.$$

Every term on the right is nonnegative. This proves Cauchy–Schwarz by finite algebra. Set $a_i = \sqrt{p_i}$ and $b_i = \sqrt{r_i}$ to obtain $1 - B(p, r)^2 \geq 0$. Equality requires $\sqrt{p_i r_j} = \sqrt{p_j r_i}$ for every pair. For probability tables, these conditions force the same normalized proportions and hence $p = r$.

This identity also makes the geometry concrete. A dot product cannot exceed the product of the lengths because the difference of their squares is a sum of nonnegative

squares. The familiar angle formula is therefore compatible with the algebra of finite lists.

7.12 The same overlap in independent repeated observations

Suppose one model assigns table p to each of N independent observations and another assigns table r. Their distributions on ordered samples have square-root overlap

$$\sum_{i_1,\dots,i_N} \sqrt{p_{i_1} r_{i_1} \cdots p_{i_N} r_{i_N}} = \left(\sum_i \sqrt{p_i r_i} \right)^N = B(p,r)^N.$$

The equality follows by expanding a product of N identical finite sums. If $p \neq r$, then $B < 1$, so this overlap decreases with sample size. The corresponding spherical distance between the full sample distributions is $2\arccos(B^N)$. More observations can make two distinct sampling models more distinguishable.

Example 7.5 (Overlap after several independent observations)**.** Take $p = (0.9, 0.1)$ and $r = (0.1, 0.9)$. A single observation has overlap 0.6. Two independent observations have overlap 0.36, and five have overlap 0.07776. The distributions being compared now live on 2^N ordered samples. We can calculate their overlap without listing those samples, because the independent-product structure factors the sum.

7.13 A numerical table near a boundary

For binary x, its arc coordinate is $2\arcsin\sqrt{x}$. At $x = 0$, 0.01, 0.04, and 0.09, these coordinates are approximately 0, 0.2003, 0.4027, and 0.6094. The corresponding square roots are 0, 0.1, 0.2, and 0.3. The table helps explain why square-root coordinates spread out small probabilities.

Example 7.6 (Two small perturbations)**.** Changing x from 0.01 to 0.02 gives spherical distance approximately 0.0834. Changing it from 0.49 to 0.50 gives distance approximately 0.0200. The first pair has the same absolute probability difference as the second but a larger square-root separation. Both comparisons are finite and use the exact angle formula.

For a statistical application, the labels and the observation procedure must be fixed before comparing tables. Adding an unused zero-probability label to both tables changes none of the overlap, Hellinger, or spherical formulas. Merging existing labels is different and can change the values. This distinction prepares the channel calculations in Chapter 10.

7.14 Exercises

7.1 Map $(1/4, 1/4, 1/2)$ to the radius-two sphere and verify its squared length.

7.2 Recover the distribution represented by $s = (1, 1, \sqrt{2})$.

7.3 Compute the overlap between $(1/2, 1/2, 0)$ and $(1/2, 0, 1/2)$.

7.4 Find the spherical distance and Hellinger distance for the pair in Exercise 3.

7.5 Prove the chord-length formula by expanding a squared norm.

7.6 Show that $B(p, r) = 0$ exactly when p and r have disjoint supports.

7.7 Show that $B(p, r) = 1$ exactly when $p = r$, using the equality condition in Cauchy–Schwarz.

7.8 Find the binary arc coordinates of $x = 0, 1/4, 1/2, 3/4, 1$.

7.9 Compare the distances from 0.01 to 0.02 and from 0.49 to 0.50 for binary probabilities.

7.10 Give two distributions whose relative entropy is infinite but whose spherical distance is strictly smaller than π.

7.11 Explain the difference between the angle α, the radius-two arc length 2α, and the chord length $4\sin(\alpha/2)$.

7.12 Write an algorithm that computes overlap and spherical distance for any two finite probability tables. Include validation of the inputs and handling of rounding near one.

7.15 Selected solutions

1. The image is $(1, 1, \sqrt{2})$, whose squared length is $1 + 1 + 2 = 4$.

3. Only the first label contributes, so $B = 1/2$.

4. The spherical distance is $2\pi/3$ and the Hellinger distance is $1/\sqrt{2}$.

6. Each summand is nonnegative. Their sum is zero exactly when every product $p_i r_i$ is zero, meaning no label has positive probability in both tables.

8. The arc coordinates are $0, \pi/3, \pi/2, 2\pi/3, \pi$.

10. For $p = (1, 0)$ and $r = (1/2, 1/2)$, $D(r\|p) = \infty$, while their spherical distance is $\pi/2$.

Laboratory investigation

1. For binary distributions $p = (1, 0)$ and $q = (1/4, 3/4)$, (a) calculate their square-root vectors on the radius-two circle, (b) find the overlap $\sum_i \sqrt{p_i q_i}$, and (c) calculate the exact Fisher–Rao distance.
2. Compare the moves $x : 0 \to 1/4$ and $x : 1/4 \to 1/2$. (a) Write both coordinate changes. (b) Compute the arc distances by $2|\arcsin\sqrt{y} - \arcsin\sqrt{x}|$. (c) Explain why equal coordinate steps have unequal lengths.

3. Take $p = (1/2, 1/2)$ and $q = (11/20, 9/20)$. (a) Calculate $\delta = q - p$. (b) Find $g_p(\delta, \delta) = \sum_i \delta_i^2 / p_i$. (c) Calculate $\sqrt{g_p(\delta, \delta)}$ and the exact spherical distance. (d) Compare numerically.
4. Take $p = (1/4, 3/4)$ and move to $q = (1/2, 1/2)$. (a) Compute the local Fisher estimate at p. (b) Compute the exact arc distance. (c) Break the move into two equal probability steps and add their local estimates. (d) Explain the effect of finer resolution.
5. Consider an edge starting at $p = (0, 1)$. (a) Explain why the expression $\sum_i \delta_i^2 / p_i$ is unavailable there for a move into the first category. (b) Compute the exact Fisher–Rao distance from $(0, 1)$ to $(1/4, 3/4)$. (c) Explain why the exact metric is useful.